



Mathematics Part 1 — Cbse Annual Exam — 80 Marks

Subject: Mathematics Part 1 | Class: Class 9

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	How many parts do the coordinate axes divide the plane into? (A) 2 (B) 6 (C) 4 (D) 8 <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	1
2	Using the identity $(a + b)^2 = a^2 + 2ab + b^2$, what is the expansion of $(5x + 2y)^2$? (A) $25x^2 + 10xy + 4y^2$ (B) $5x^2 + 20xy + 2y^2$ (C) $5x^2 + 10xy + 2y^2$ (D) $25x^2 + 20xy + 4y^2$ <i>[Ch 4: Exploring Algebraic Identities]</i>	1
3	Two coins are tossed at the same time. What is the probability of getting at least one head? (A) $1/4$ (B) $3/4$ (C) $1/2$ (D) 1 <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	1



Q.NO.	QUESTIONS	MARKS
4	Assertion (A): 471 is not a term of the sequence given by the formula $s_n = 5n - 2$. Reason (R): For a number to be a term of the sequence, the value of n must be a natural number. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]</i>	1
5	By the Baudhāyana-Pythagoras Theorem, what is the length of the diagonal of a square where each side is exactly 1 unit long? (A) $\sqrt{2}$ (B) 2 (C) 1 (D) $\sqrt{3}$ <i>[Ch 3: The World of Numbers]</i>	1
6	A line drawn from the centre of a circle to the midpoint of a chord is always: (A) Perpendicular to the chord (B) Parallel to the chord (C) Equal to the chord (D) A diameter of the circle <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	1
7	What value must the fourth side d of a 4-gon take for Brahmagupta's formula to reduce to Heron's formula? (A) 1 (B) a (C) -1 (D) 0 <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	1
8	What is the simplified fractional form of the pure repeating decimal $0.4\bar{5}$? (A) $45/100$ (B) $9/20$ (C) $5/11$ (D) $15/90$ <i>[Ch 3: The World of Numbers]</i>	1



Q.NO.	QUESTIONS	MARKS
9	The area of a sector of a circle with radius r that subtends an angle θ° at the centre is: (A) $\pi r^2 \times \theta/180$ (B) $\pi r^2 \times \theta/360$ (C) $2\pi r \times \theta/360$ (D) $\pi r^2 \times \theta/90$ <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	1
10	In the linear relationship $y = ax + b$, what does the constant b represent? (A) The slope of the line (B) The x-intercept (C) The degree of the polynomial (D) The y-intercept, which is the distance from the origin where the line cuts the y-axis <i>[Ch 2: Introduction to Linear Polynomials]</i>	1
11	What does probability measure? (A) Physical quantities like length and volume (B) The likelihood of events (C) The exact outcome of an experiment (D) The speed of moving objects <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	1
12	Assertion (A): When triangle ADM is reflected in the y-axis to form triangle A'D'M', the length A'D' equals the length AD. Reason (R): The distance between two points is given by $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, and it makes no difference whether $(x_2 - x_1)$ and $(y_2 - y_1)$ are positive or negative. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	1
13	Which Greek mathematician described the latitudes and longitudes of thousands of locations, including 'Ozine' (Ujjayinī)? (A) Hipparchus (B) Āryabhaṭa (C) Ptolemy (D) Baudhāyana <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	1



Q.NO.	QUESTIONS	MARKS
14	Assertion (A): The Virahāṅka-Fibonacci sequence is an arithmetic progression. Reason (R): In an arithmetic progression, the difference between consecutive terms is constant. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]</i>	1
15	If two opposite angles of a quadrilateral add up to 180°, what can be concluded about its vertices? (A) They form a parallelogram (B) They are collinear (C) They lie on a circle (are concyclic) (D) They are equidistant from the centre <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	1
16	In the equation $y = ax + b$, what does a represent? (A) Slope of the line (B) y-intercept of the line (C) x-intercept of the line (D) Origin <i>[Ch 2: Introduction to Linear Polynomials]</i>	1
17	Which of the following identities represents the factorization of $x^3 - y^3$? (A) $(x - y)(x^2 - xy + y^2)$ (B) $(x - y)(x^2 + xy + y^2)$ (C) $(x + y)(x^2 - xy + y^2)$ (D) $(x + y)(x^2 + xy + y^2)$ <i>[Ch 4: Exploring Algebraic Identities]</i>	1
18	Assertion (A): In a random experiment like tossing a coin, you cannot predict exactly what will happen. Reason (R): Randomness refers to a situation where you may know all the possible outcomes, but you cannot say which one will definitely occur. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	1



Q.NO.	QUESTIONS	MARKS
19	The nth term of the odd number sequence is given by $u_n = 2n - 1$. Which position does the number 137 occupy in this sequence? (A) 67 (B) 69 (C) 68 (D) 70 <i>[Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]</i>	1
20	Assertion (A): The product $(x-y)(x^2+xy+y^2)$ simplifies to $x^3 + y^3$. Reason (R): When $(x-y)$ is multiplied by (x^2+xy+y^2) using the distributive property, the terms x^2y and xy^2 cancel out. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 4: Exploring Algebraic Identities]</i>	1
Section B Attempt 5 out of 7.		
21	Why is π classified as an irrational number? <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	2
	OR Describe the construction steps to locate $\sqrt{2}$ on the number line using a unit square's diagonal. <i>[Ch 3: The World of Numbers]</i>	
22	Find the value of 119^2 by expressing 119 as $100 + 10 + 9$ and applying a suitable identity. <i>[Ch 4: Exploring Algebraic Identities]</i>	2
	OR A chord of length 24 cm is drawn in a circle of diameter 26 cm. Find the distance from the centre of the circle to the chord. <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	
23	In a survey of 50 students, 15 students said they liked football. The number of students who like football is 15, and the ____ (frequency/relative frequency) is ____ (fill in the fraction or decimal). <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	2
24	What role did Ujjayinī play in the ancient world regarding coordinates? <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	2



Q.NO.	QUESTIONS	MARKS
25	An auto-rickshaw fare is ₹25 for the initial 2 km and increases by ₹15 per km thereafter. Write the simplified linear expression for the total fare for n km, where $n \geq 2$. <i>[Ch 2: Introduction to Linear Polynomials]</i>	2
Section C Attempt 5 out of 7.		
26	The number $0.\bar{9}$ (which means 0.99999 ...) is a rational number. Using algebra (let $x = 0.\bar{9}$, multiply by 10, and subtract), explain why $0.\bar{9}$ is exactly equal to 1. <i>[Ch 3: The World of Numbers]</i>	3
	OR Determine the AP whose third term is 16 and whose 7th term exceeds the 5th term by 12. <i>[Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]</i>	
27	Solve the previous question using the Baudhāyana–Pythagoras theorem. <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	3
	OR Find the value of the quadratic polynomial $7s^2 - 4s + 6$ when $s = -3$. <i>[Ch 2: Introduction to Linear Polynomials]</i>	
28	Define a segment of a circle. A chord of a circle of radius 10 cm subtends 90° at the centre. Find the area of the corresponding minor sector. (Use $\pi \approx 3.14$) <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	3
29	What is a random experiment in probability? Give one example and explain why it qualifies as random. <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	3
30	State the general formula used to find the distance between two points (x_1, y_1) and (x_2, y_2) in the 2-D plane. Why does it not matter whether $(x_2 - x_1)$ and $(y_2 - y_1)$ are positive or negative? <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	3
Section D Attempt 4 out of 6.		
31	Suppose a plant has height 1.75 feet and it grows by 0.5 feet each month. (i) Find the height after 7 months. (ii) Make a table of values for t varying from 0 to 10 months and show how the height, h, increases every month. (iii) Find an expression that relates h and t, and explain why it represents linear growth.	5



Q.NO.	QUESTIONS	MARKS
	<i>[Ch 2: Introduction to Linear Polynomials]</i>	
	OR A city has two main roads which cross each other at the centre of the city. These two roads are along the North-South (N-S) direction and East-West (E-W) direction. All the other streets of the city run parallel to these roads and are 200 m apart. There are 10 streets in each direction. (i) Using 1 cm = 200 m, draw a model of the city in your notebook. Represent the roads/streets by single lines. (ii) There are street intersections in the model. Each street intersection is formed by two streets—one running in the N-S direction and another in the E-W direction. Each street intersection is referred to in the following manner: If the second street running in the N-S direction and 5th street in the E-W direction meet at some crossing, then we call this street intersection (2, 5). Using this convention, find: (a) how many street intersections can be referred to as (4, 3). (b) how many street intersections can be referred to as (3, 4). <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	
32	Prove that the opposite angles of a cyclic quadrilateral add up to 180°. Explain how the angles subtended by arcs at the centre and at the circumference are used in this proof. <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	5
	OR State the identity for $(a + b + c)^2$ and describe its geometric interpretation using a square of side $(a + b + c)$. Use this identity to find the value of 119^2 and to expand $(3x - 2y + 4z)^2$. <i>[Ch 4: Exploring Algebraic Identities]</i>	
33	Find 5 rational numbers between $\frac{2}{5}$ and $\frac{3}{5}$. Show that the rational number $\frac{(a+b)}{2}$ always lies between any two rational numbers a and b. Using this result, explain what is meant by the density of rational numbers on the number line. <i>[Ch 3: The World of Numbers]</i>	5
34	A triangle has side lengths of 3 units, 4 units, and 5 units. Calculate its area using Heron's formula. Then, verify the area using the 'half base times height' formula, justifying why the triangle is right-angled. <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	5
Section E Attempt 3 out of 4.		



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35	Case Study: City Map Coordinates A student is creating a 2-D map of a city center, inspired by how early Arabic maps used the Ujjayinī meridian as a zero-longitude reference. On the map, the main intersection is chosen as the point of intersection of the axes, called the origin. The horizontal road is the x-axis and the vertical road is the y-axis. A cafe is located 3 units to the right and 2 units upwards from the origin. A library is located 4 units to the left and 5 units upwards from the origin. A park is located 2 units to the left and 3 units downwards from the origin. The student marks these locations using coordinates. (i) What are the coordinates of the origin? (ii) The park is located 2 units to the left and 3 units downwards from the origin. In which quadrant does the park lie? (iii) The cafe is at (3, 2). A shop is located at (7, 2). Find the distance between the cafe and the shop. (iv) A bookstore is located on the y-axis, 6 units upwards from the origin, and a florist is located on the x-axis, 6 units to the right of the origin. Are the coordinates of the bookstore and the florist the same? Explain. <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	5
	OR Case Study: Algebraic Expressions and Terms A shop has sealed boxes of different colours for sale. Red boxes have 4 pens each and blue boxes have 5 pencils each. If a customer buys x red boxes and y blue boxes and gets 3 extra pens free, the total quantity of pens and pencils is given by the algebraic expression $4x + 5y + 3$. In this expression, $4x$, $5y$ and 3 are terms, x and y are variables, the numbers 4 and 5 are the coefficients of x and y respectively, and 3 is a constant. (i) What are the variables used in the expression $4x + 5y + 3$? (ii) What is the coefficient of y in the given expression? (iii) Identify the terms and the constant in the expression $4x + 5y + 3$. (iv) If a customer buys 2 red boxes and 1 blue box, what is the total quantity of pens and pencils they get? <i>[Ch 2: Introduction to Linear Polynomials]</i>	



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36	Case Study: Finding Squares Using Identities A student was preparing for a mental math competition. She learned that identities simplify calculations significantly. To find 29^2, she wrote $(30 - 1)^2$ and used the identity $(a - b)^2 = a^2 - 2ab + b^2$ to get $900 - 60 + 1 = 841$. To find 119^2, she expressed it as $(100 + 10 + 9)^2$ and applied the identity $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$. She also learned that Śhrīdharāchārya in 750 CE proposed the identity $a^2 = (a + b)(a - b) + b^2$, which she used to find $55^2 = (55 + 5)(55 - 5) + 5^2 = 60 \times 50 + 25 = 3025$. (i) Using the identity $(a - b)^2 = a^2 - 2ab + b^2$, find the value of 79^2 by expressing 79 as $(80 - 1)$. (ii) To calculate 97^2, a student writes it as $(100 - 3)^2$. Which identity should she apply, and what will be the expanded form? (iii) Using the identity $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$, find the value of 117^2 by expressing 117 as $(100 + 10 + 7)$. (iv) Using Śhrīdharāchārya's method $a^2 = (a + b)(a - b) + b^2$, calculate 35^2 by choosing $b = 5$. Explain why $b = 5$ is a suitable choice. <i>[Ch 4: Exploring Algebraic Identities]</i>	5
37	Case Study: Cyclic Quadrilateral Park A city park is shaped like a cyclic quadrilateral ABCD, with a circular walking path forming its boundary. The angles at the corners are carefully measured to ensure the design is accurate. The angle at corner A is 75 degrees, and the angle at corner B is 110 degrees. The park designer uses the properties of cyclic quadrilaterals to verify the layout. One such property states that opposite angles of a cyclic quadrilateral add up to 180 degrees. The converse property is also considered for future designs. (i) What is the measure of angle C? (ii) What is the measure of angle D? (iii) If angle A was $(2x + 10)$ degrees and angle C was $(3x - 20)$ degrees, find the value of x. (iv) State the converse property that determines if a quadrilateral is cyclic based on its opposite angles. <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	5

